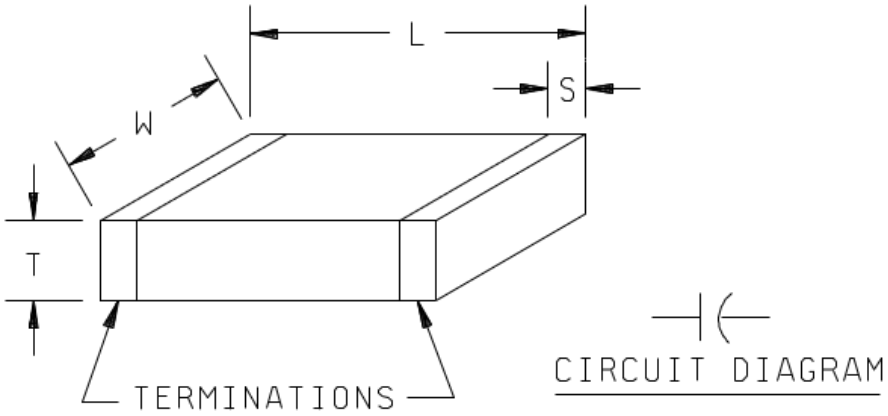
INCH-POUND
MIL-PRF-123/12F
3 June 2026
SUPERSEDING
MIL-PRF-123/12E
w/AMENDMENT 1
5 November 2020

PERFORMANCE SPECIFICATION SHEET

CAPACITOR, FIXED, CERAMIC DIELECTRIC (TEMPERATURE STABLE AND GENERAL PURPOSE), HIGH RELIABILITY, NONLEADED, STYLE CKS53

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-123.



Inches	mm
.010	0.25
.015	0.38
.020	0.51
.065	1.65
.080	2.03
.180	4.57

L	W	T		S
± .015	± .015	Min	Max	± .010
.180	.080	.020	.065	.020

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in millimeters and are given for general information only.
3. Dimensions and tolerances are for bare chips. For solder coated terminations, add .025 inch (0.64 mm) to the positive length (L) tolerance, .015 inch to the positive width (W) and thickness (T) tolerances, and .0125 inch (0.32 mm) to the positive S dimension tolerance.

FIGURE 1. Style CKS53 capacitors.

AMSC N/A
FSC 5910
DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:13Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Multilayer, unencapsulated, monolithic ceramic.

Product level: A or T in accordance with MIL-PRF-123.

Termination: M, G, S, or Z in accordance with MIL-PRF-123.

Capacitance value: See table I.

Capacitance tolerance: See table I. (F = ± 1 percent; J = ± 5 percent; K = ± 10 percent.)

Voltage-temperature limit: BP and BX in accordance with MIL-PRF-123.

Operating temperature: -55°C to +125°C.

Rated voltage: See table I (B = 50 Vdc, C = 100 Vdc).

Marking: In accordance with MIL-PRF-123.

TABLE I. Style CKS53 characteristics.

Part or Identifying Number (PIN) 1/	Capacitance (pF)	Capacitance tolerance	Voltage-temperature limits	Rated voltage (V dc)
M123 - 12BP - 301 - -	300	F, J, K	BP	50, 100
M123 - 12BP - 331 - -	330	F, J, K	BP	50, 100
M123 - 12BP - 361 - -	360	F, J, K	BP	50, 100
M123 - 12BP - 391 - -	390	F, J, K	BP	50, 100
M123 - 12BP - 431 - -	430	F, J, K	BP	50, 100
M123 - 12BP - 471 - -	470	F, J, K	BP	50, 100
M123 - 12BP - 511 - -	510	F, J, K	BP	50, 100
M123 - 12BP - 561 - -	560	F, J, K	BP	50, 100
M123 - 12BP - 621 - -	620	F, J, K	BP	50, 100
M123 - 12BP - 681 - -	680	F, J, K	BP	50, 100
M123 - 12BP - 751 - -	750	F, J, K	BP	50, 100
M123 - 12BP - 821 - -	820	F, J, K	BP	50, 100
M123 - 12BP - 911 - -	910	F, J, K	BP	50, 100
M123 - 12BP - 102 - -	1,000	F, J, K	BP	50, 100
M123 - 12BX - 562K -	5,600	K	BX	50, 100
M123 - 12BX - 682K -	6,800	K	BX	50, 100
M123 - 12BX - 822K -	8,200	K	BX	50, 100
M123 - 12BX - 103K -	10,000	K	BX	50, 100
M123 - 12BX - 123K -	12,000	K	BX	50, 100
M123 - 12BX - 153K -	15,000	K	BX	50, 100
M123 - 12BX - 183K -	18,000	K	BX	50, 100
M123 - 12BX - 223K -	22,000	K	BX	50, 100
M123 - 12BX - 273K -	27,000	K	BX	50, 100
M123 - 12BX - 333K -	33,000	K	BX	50, 100

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:13Z
Check the source to verify that this is the current version before use.

TABLE I. Style CKS53 characteristics - Continued.

PIN 1/	Capacitance (pF)	Capacitance tolerance	Voltage-temperature limits	Rated voltage (V dc)
M123 - 12BXB393K -	39,000	K	BX	50
M123 - 12BXB473K -	47,000	K	BX	50
M123 - 12BXB563K -	56,000	K	BX	50
M123 - 12BXB683K -	68,000	K	BX	50
M123 - 12BXB823K -	82,000	K	BX	50
M123 - 12BXB104K -	100,000	K	BX	50

1/ The complete PIN will include additional letter(s) to indicate product level, voltage, capacitance tolerance, and termination, as applicable.

Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Navy - EC
Air Force - 19
DLA - CC
Other – NA

Preparing activity:
DLA - CC

(Project 5910-2026-023)

Review activity:
Air Force - 85

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:13Z
Check the source to verify that this is the current version before use.